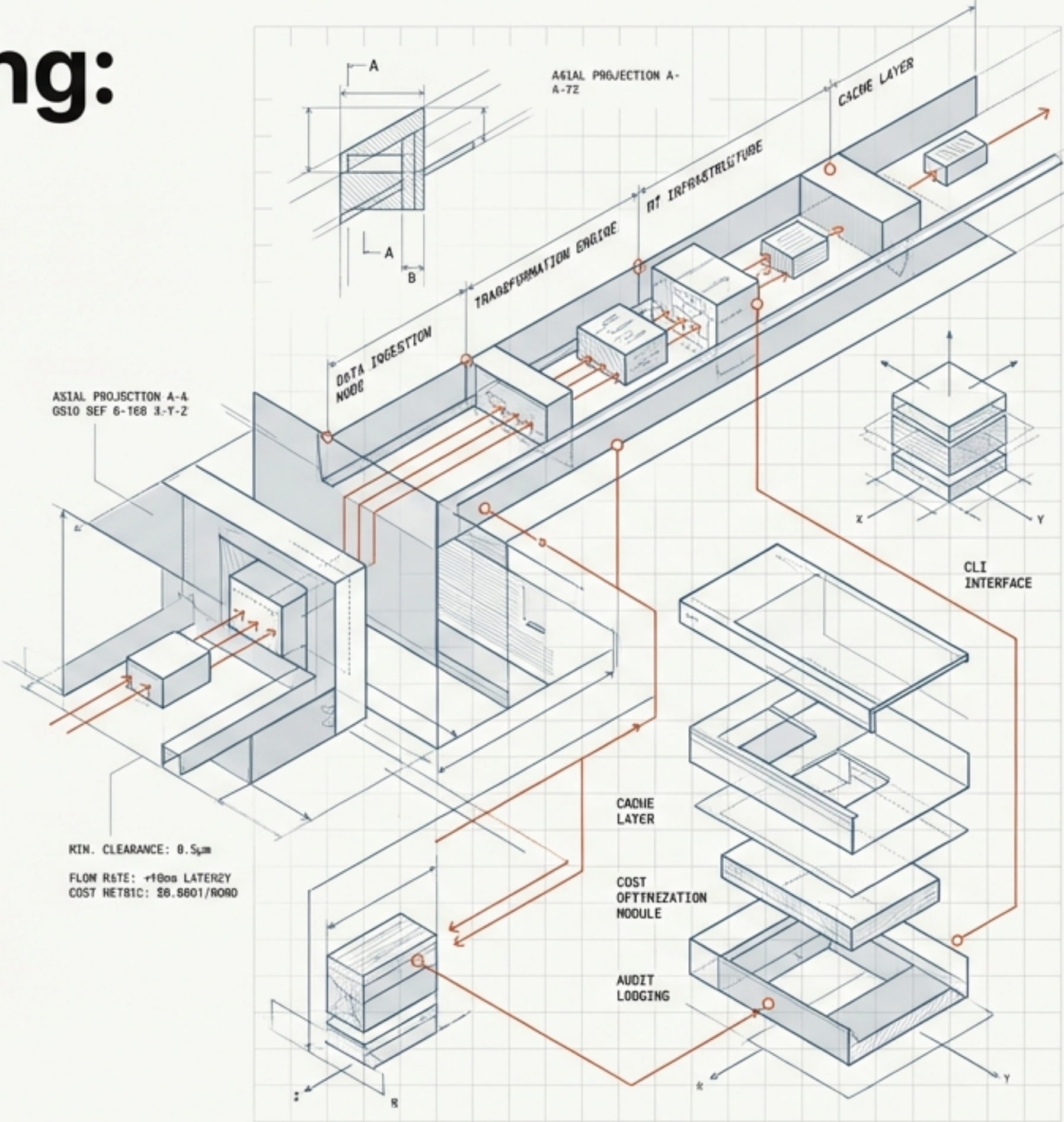


Translation Engineering: Infrastructure, Cost Optimisation, and CLI Architecture

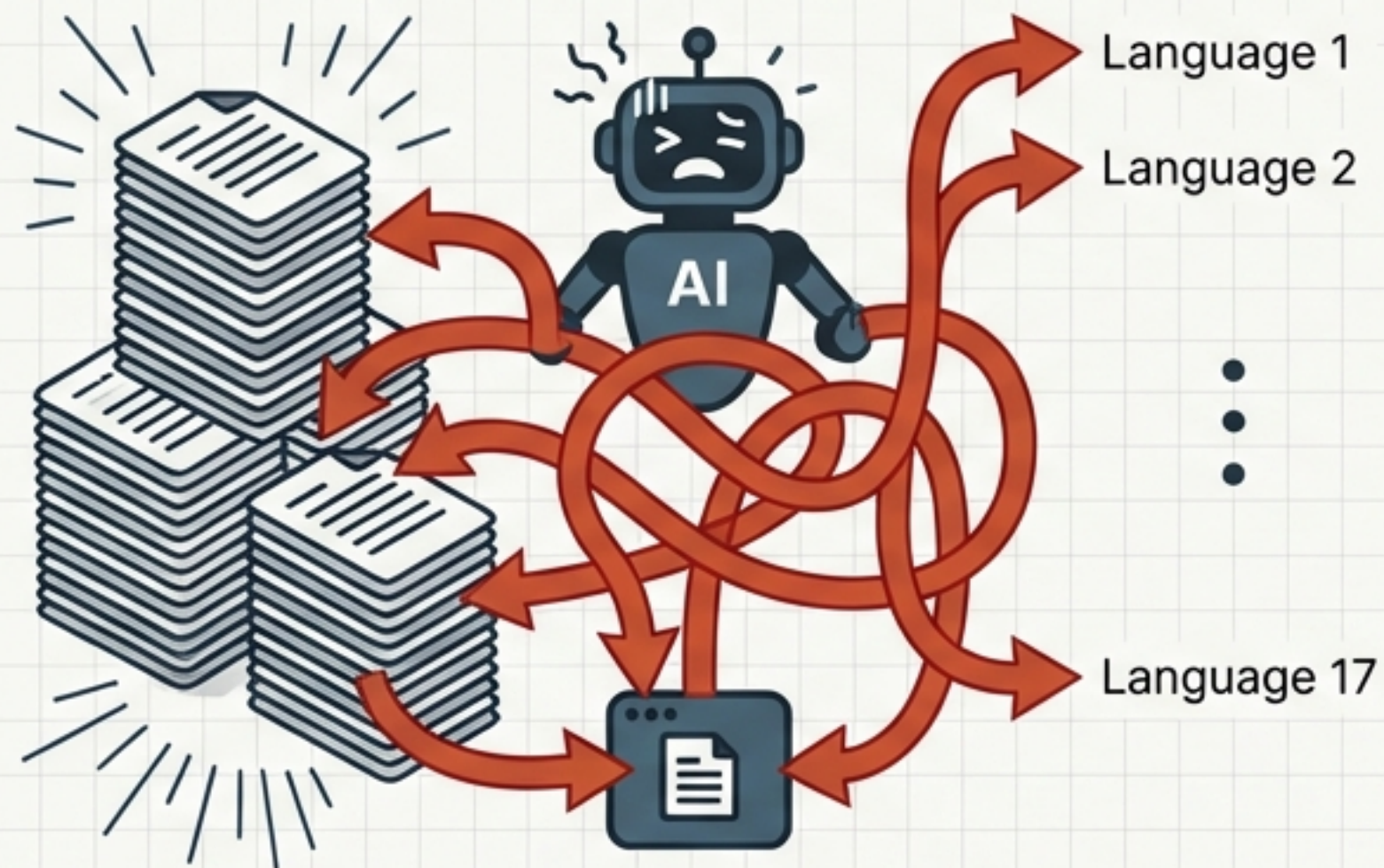


Version:	v0.12.2
Date:	7 Mar 2026
To:	Technical Leadership, Lead Developers, Software Architects, Accountants



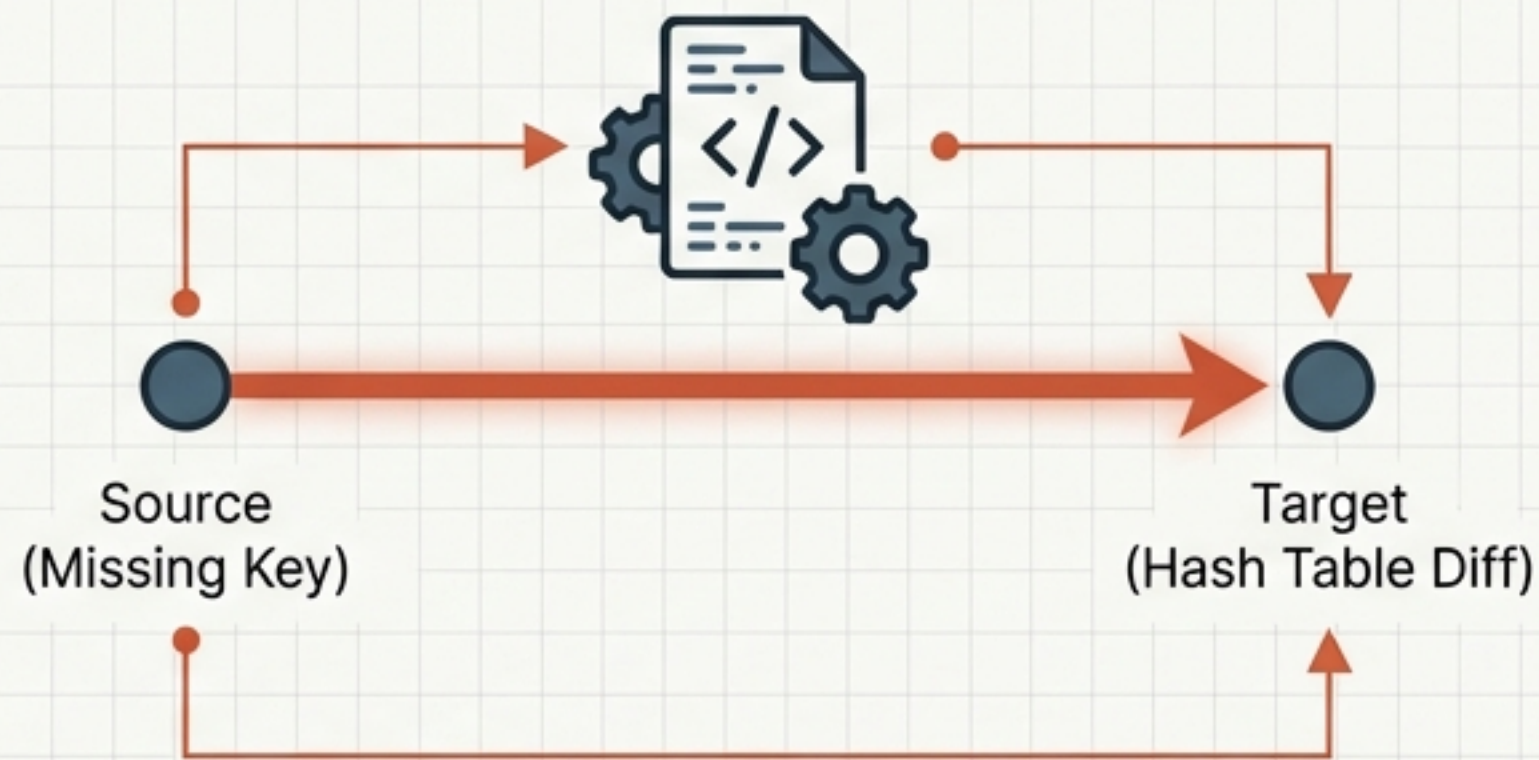
The LLM Cost Engineering Problem: Translations Do Not Scale

The Absurdity



17 Languages = ~34 full files read/written just to add one key.

The Reality

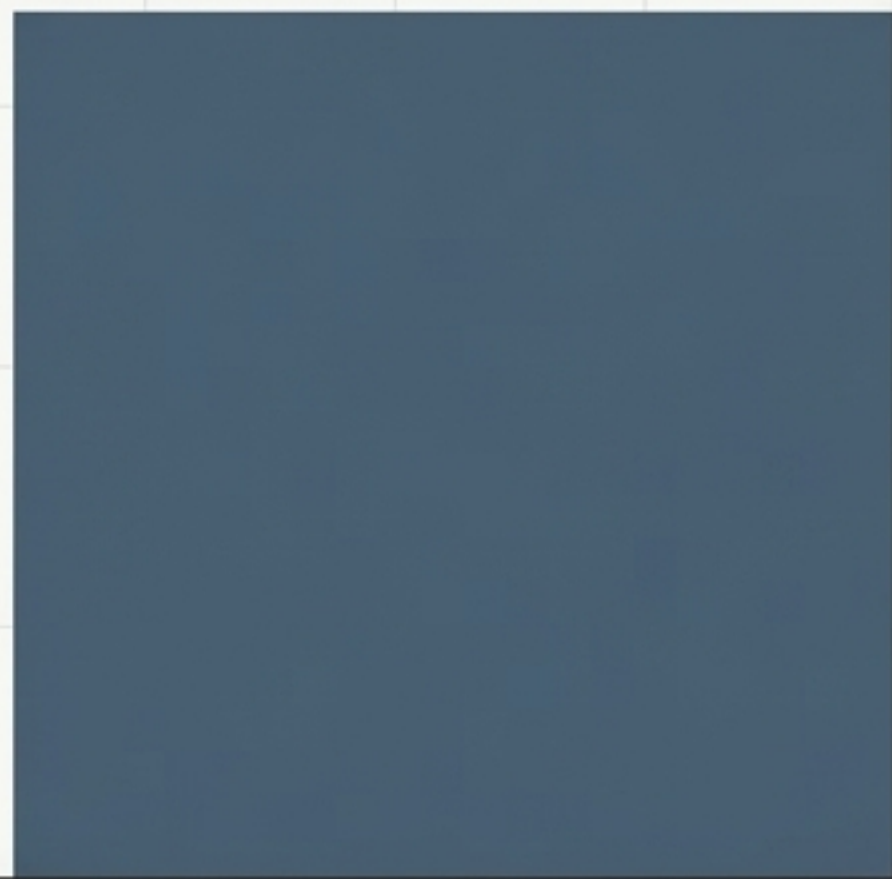


This is a hash table diff. It takes milliseconds and zero tokens.

Adding a single missing key currently forces an LLM agent to read and write thousands of tokens across 17 languages. This is catastrophically token-inefficient.

Use the LLM to write the script, not to do the work.

~20K tokens (reading full files)

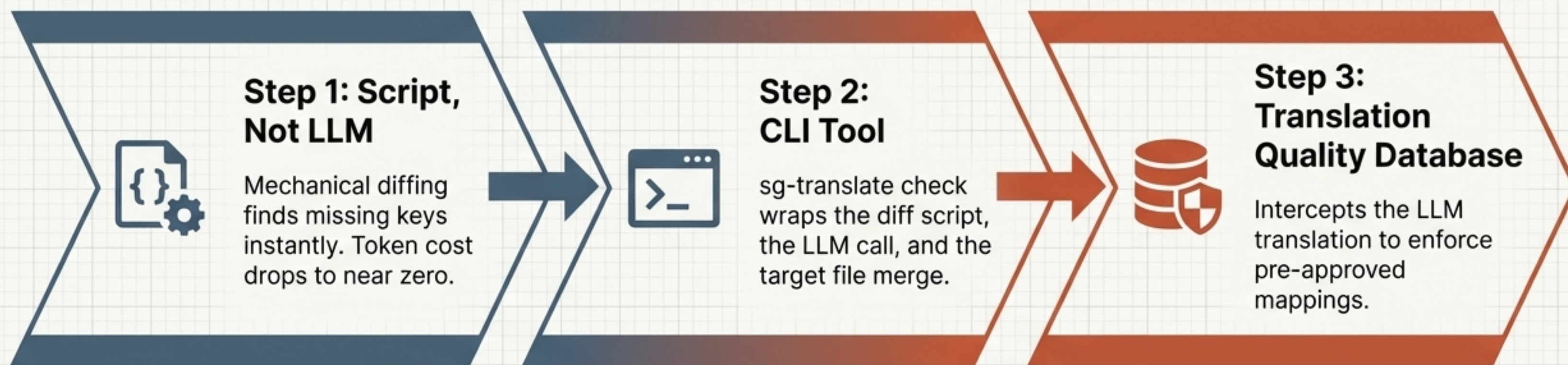


~500 tokens
(translating missing keys)



The transition from manual agent reading to mechanical diffing yields an immediate 40x cost reduction per language.

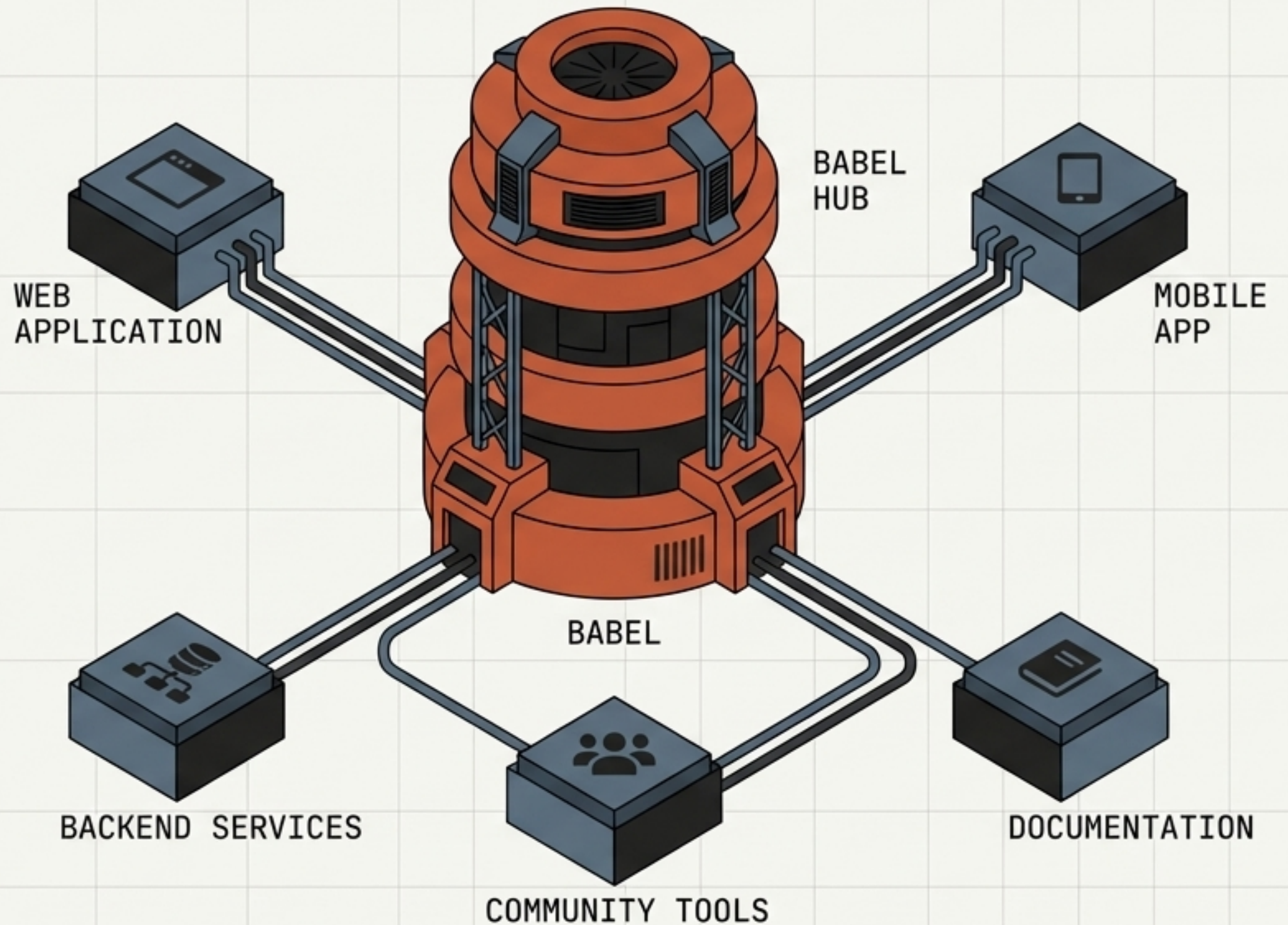
The Engineered Translation Pipeline



The transition from manual agent reading to mechanical diffing yields an immediate 40x cost reduction per language.

Introducing babel.sgraph.ai

A standalone platform serving as the definitive hub for all translations. It provides a universal translation API and enables seamless community contribution across our entire ecosystem.



Core Platform Capabilities



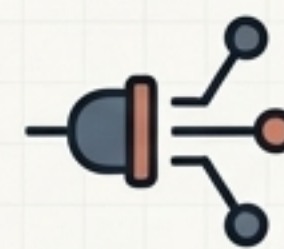
Translation Registry

All key-value pairs, hashes, and verification statuses.



Shared Vocabulary

Pre-approved terms for new projects to query.



Translation API

Runtime consumption endpoint for all SG/Send projects.



Community UI

Pull request workflows for human translators.

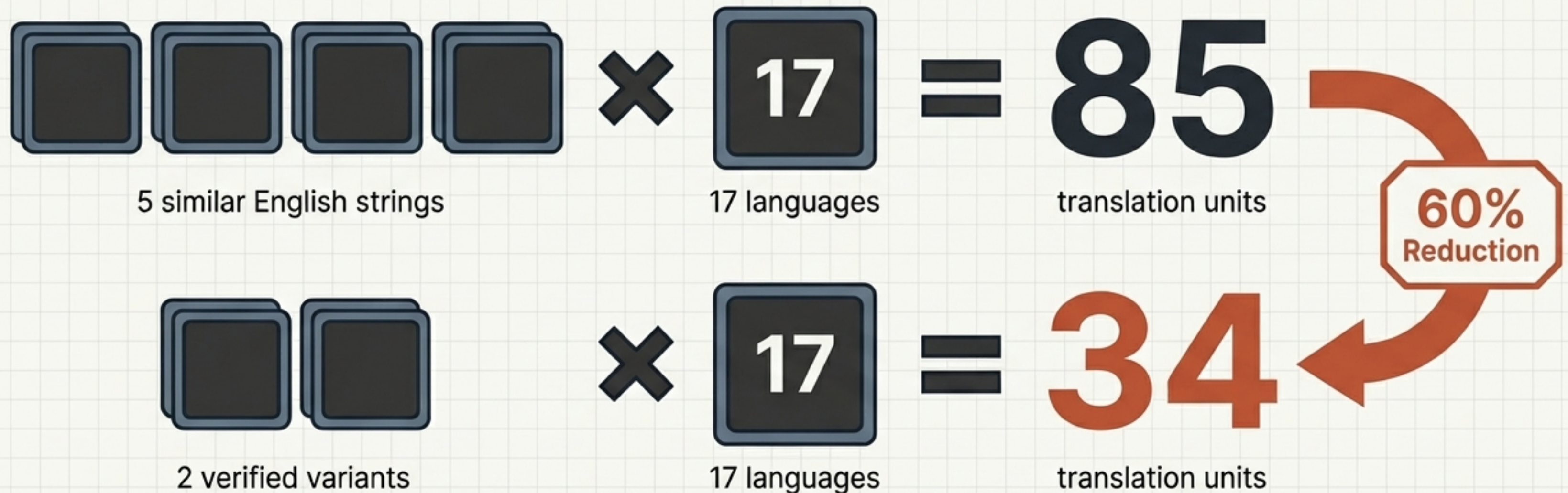


Quality Dashboard

Real-time metrics on coverage and known issues.

The Economics of Shared Vocabulary

Enforcing a shared vocabulary before creating new text results in a **60% reduction in ongoing translation costs**. New variants are allowed, but the cost multiplier is made visible.



Enforcing a shared vocabulary before creating new text results in a **60% reduction in ongoing translation costs**. New variants are allowed, but the cost multiplier is made visible.

The <sg-translations> Web Component

Components no longer manage their own translation files. They simply request keys from the central service via events.

sg-layout

Visible UI Panels

UI Panel A

UI Panel B

UI Panel C

...

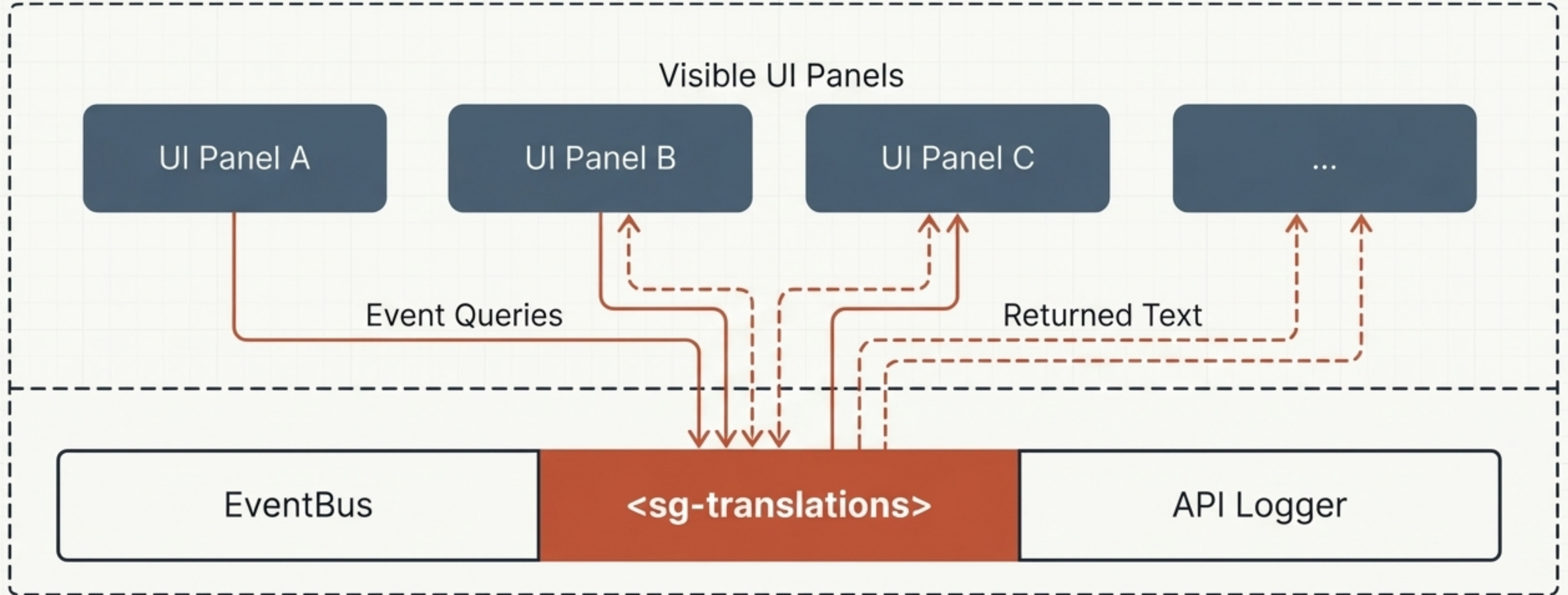
Event Queries

Returned Text

EventBus

<sg-translations>

API Logger



Component Superpowers



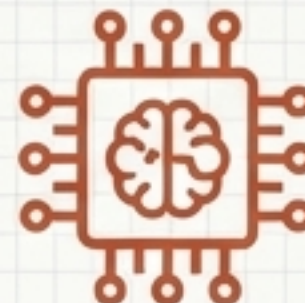
Aggressive Caching

Uses localStorage and in-memory caching for instant, synchronous responses.



Zero-Code Scaling

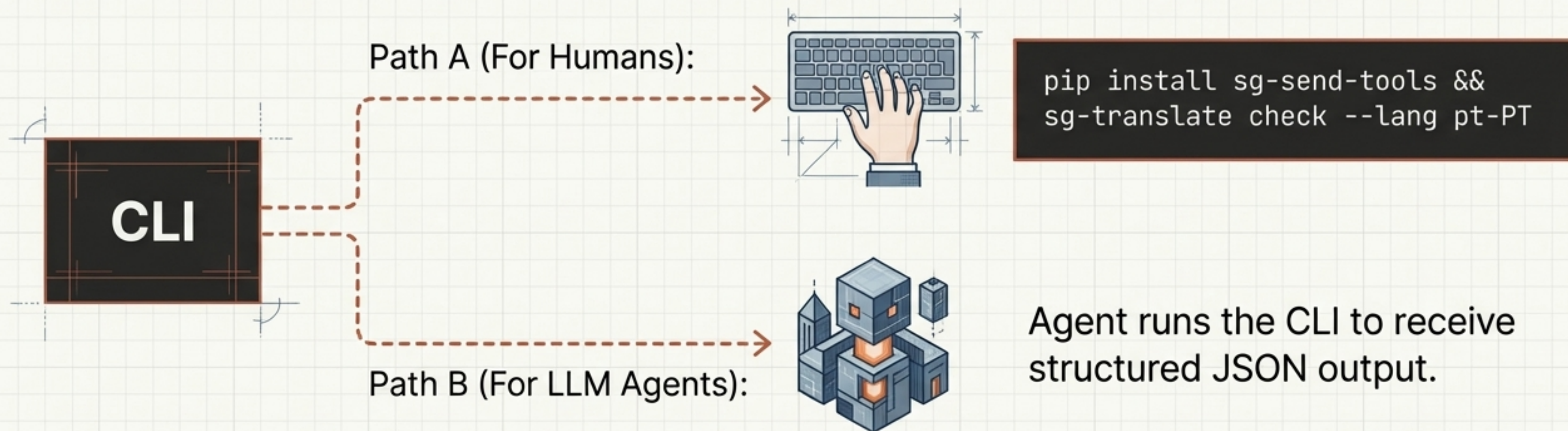
Adding a new language simply means loading new data—no changes to component code required.



Local LLM Ready

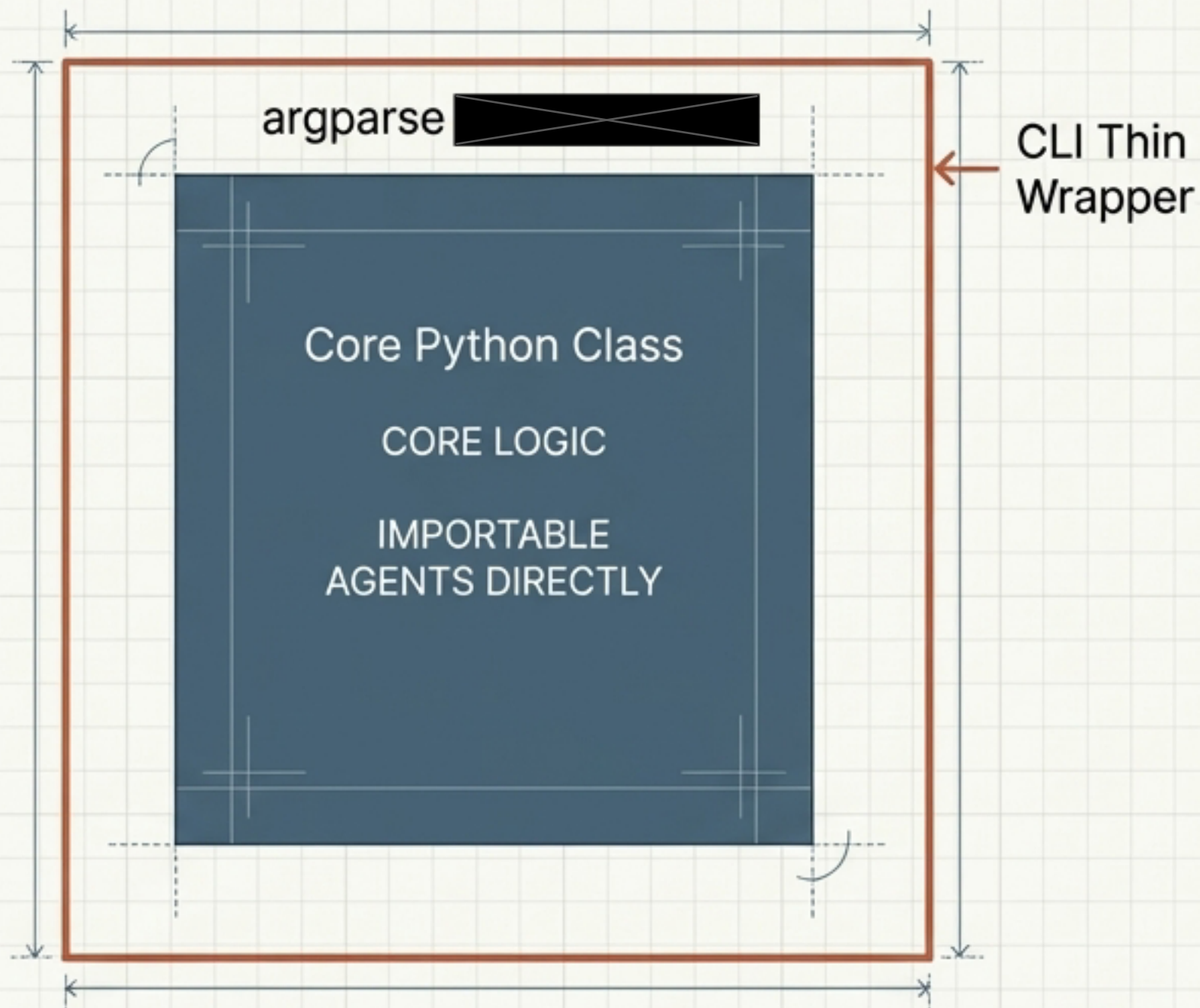
Architecture perfectly supports on-the-fly local LLM runtime translation for unsupported languages.

Fractal CLIs: Every Project Gets a Tool



Use LLMs to not use LLMs. The CLI does the mechanical work; the agent consumes the output to make decisions.

CLI Architecture & Standards



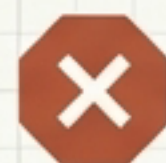
Builds on the established class-based pattern from the vault CLI.



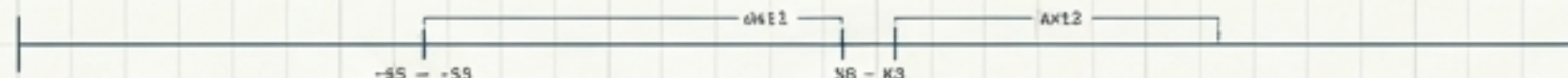
Core classes must be importable for use in other code or by agents directly.



CLI remains a thin wrapper using argparse.



Strict Constraint: No dependency on Typer. Keep it consistent.



Redefining Governance: The Accountant

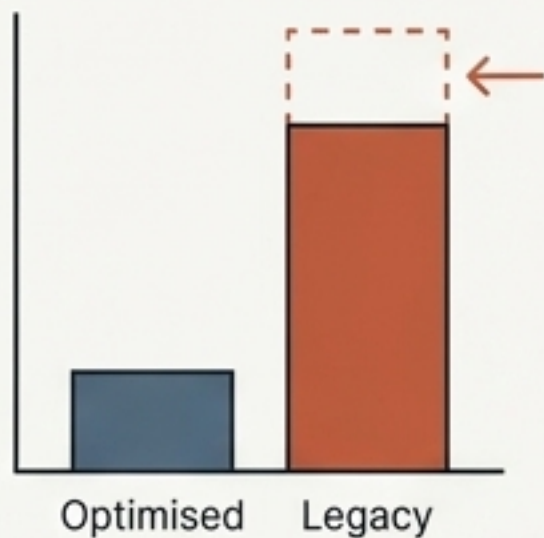


**Tokens are a bellwether
for engineering quality.**

If a task consumes excessive tokens, it is either genuinely complex, or the workflow is unoptimised. The Accountant's role is to distinguish between the two.

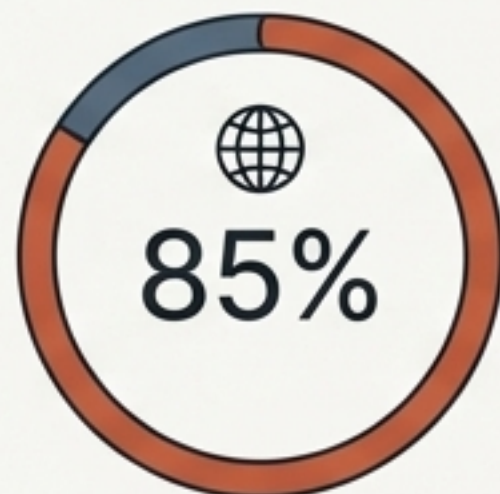
The Accountability Dashboard

Tokens per update



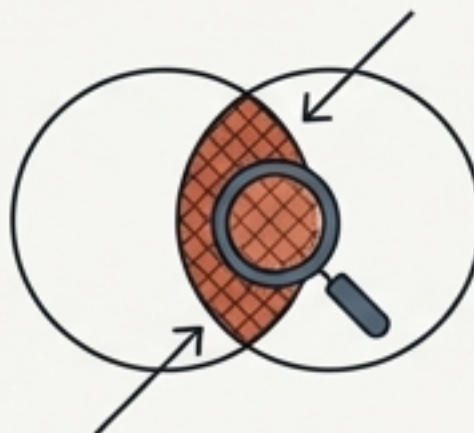
Optimised pipeline vs. legacy workflow.

Coverage



Percentage of keys translated per language.

Duplication



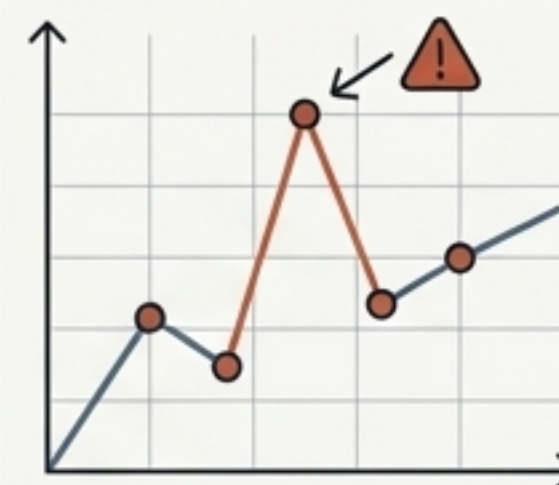
Detection of identical strings for consolidation.

Cost per addition



True cost of 1 new English key $\times$ N translations.

Session usage



Tracking session tokens to flag poor engineering practices.

Execution & Acceptance Criteria

Pipeline	Babel Platform	Web Component
☒ Script-based diff implemented	☒ Registry live with hashes/status	☒ <sg-translations> providing events
☒ LLM translates only missing keys	☒ Shared vocabulary enforced	☒ localStorage caching working
☒ sg-translate CLI functional	☒ API endpoints active	☒ English fallback active
☒ Quality database active	☒ Community UI functional	☒ Deployed within sg-layout (JetBrains for sg-layout)

Document Control

Translation Engineering Brief — babel.sgraph.ai
v0.12.2 // End of Brief

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